

bthome-writable — one page for the BTHome maintainers

A briefing, not a draft: facts and links to speak from. The submission is the owner's own words — the Open Home Foundation [AI policy](#) closes contributions believed to be agent-written (`decisions.md` D-091).

The ask

Reserve one object ID: 0xFF . Everything else reuses BTHome's own object table, encodings and AES-CCM. And a question rather than a request: the specification is BTHome's to take if the maintainers want it.

What it does

A device lists, in its ordinary BTHome advertising, the object types it accepts writes for — object 0xFF , **last in the service data**, so existing receivers skip it. A receiver connects briefly and writes **one BTHome object** to that entry's own GATT characteristic. The write response is the acknowledgement; a writable value is never advertised.

```
advertising  40 00 09 01 61 FF 01 1E      battery, and "entry 1 accepts 0x1E"
write        1E 01  -> 2FAA0001          one object, one characteristic
```

It answers requests already made here

- [bthome-ble #257](#) — "BTHome should define an Object Id for supported events": a declaration object, asked for independently in 2025 for a narrower case.
- [#146](#) two-way communication — Ernst79: "we will welcome contributions from others if they want to add this somehow." [#287](#) controlling Shelly BLU devices.

Re-checked 2026-10-07: no downlink proposed anywhere, and 0xFF unassigned — 0xF2 is the highest ID in use.

Evidence

Three independent sites. Gordon Williams on a Puck.js: "works great", connections brief, no missed writes. @enaon on an RPi4b with OpenWrt, six ESPHome proxies and local BlueZ: working over both paths, and his report found a fault the original bench could not produce.

Measured through Home Assistant	
Click to action, 1 s advertising interval	1.7 s , of which 36 ms is the write
Repeated commands	0.31 s
Delivered, two devices	100 % , no retries
Declaration on the air, three writable lights	5 bytes

Two reference implementations, a shared test-vector contract with AES-CCM vectors that pass on-device, 683 automated tests across three suites.

Objections, and the short answers

- **Slow, and costly in battery.** 1.7 s click to action. The idle advertising interval is the device application's choice and is unchanged by this.
- **Unknown objects break existing receivers.** bthome-ble skips an unknown ID and stops parsing, so the declaration goes last and existing installs lose nothing. Verified against the real parser in CI.
- **Writes are a security hole.** BTHome's own AES-CCM both directions, the direction bound into the nonce so no recording replays as another, and a monotonic write counter. Without the key there is no write.

Links

<https://github.com/yerpj/bthome-writable> — specification `spec/PROTOCOL.md` , every resolved question with the measurement that resolved it `spec/decisions.md` , ten-minute replication `docs/try-it.md` . Design converged with Gordon Williams in [espruino#8024](#).